IRexp: A database of experimental infrared band lists from open literature

Ilkham Yabbarov[1*] and Rodrigo A. Vargas-Hernández[1,2,3*]

[1*]Department of Chemistry and Chemical Biology, McMaster University, Hamilton, Ontario, L8S 4L8, Canada.
[2]Brockhouse Institute for Materials Research, McMaster University, Hamilton, Ontario, L8S 4L8, Canada.
[3]School of Computational Science and Engineering, McMaster University, Hamilton, Ontario, L8S 4L8, Canada.

*Corresponding author(s). E-mail(s): yabbaroi@mcmaster.ca; vargashr@mcmaster.ca;

Abstract

IRexp is a redistributable collection of experimental infrared band lists (cm^{-1} peak positions) mined from open chemistry literature, optionally with author-reported $^{1}H/^{13}C$ NMR strings and resolved structures. The release holds 121,233 records (119,345 PMC Open Access Subset; 1,888 Chemotion/RADAR4Chem), including 43,060 structure-linked entries and 33,201 full IR + ^{1}H + ^{13}C + structure quadruples. IRexp stores numeric band lists, not absorbance traces; each record carries a source DOI and stamped licence / licence-pool fields. Licensing is mixed: after Europe PMC joins plus conservative Crossref recovery of previously empty licences, 88,545 records are commercial (CC-BY/CC0), 21,823 non-commercial (NC*), 8,963 empty/unknown (excluded from the commercial deposit), 1,897 ShareAlike (Chemotion plus rare PMC SA), and 5 other (ND). The resource is intended for multimodal training, retrieval, and tool-input in spectroscopic agents and autonomous chemistry workflows that need structured, attributable experimental peak lists rather than paywalled PDFs or view-only archives. Technical validation covers automated transcription, harvest-path recall proxies, stratified chemist-proxy checks, and full-corpus quarantine; it does not claim NMRexp-equivalent human expert audits. Complementary elucidation benchmarks are described in a companion research manuscript and are not analysed here.

Keywords: infrared spectroscopy, band lists, peak lists, PMC Open Access, Chemotion, FAIR data, licence pools, data descriptor

1 Background & Summary

Infrared (IR) spectroscopy is routine in organic characterisation, yet open, redistributable collections of *experimental* IR data remain sparse relative to modern machine-learning and agentic tool-use needs. Digitised absorbance libraries such as the NIST Chemistry WebBook [1] and AIST SDBS [2] are valuable but either modest in size or view-only (no bulk redistribution). Computational IR–NMR resources published in *Scientific Data* expand multimodal coverage with simulated spectra [3], while large literature mines for NMR peak lists (notably NMRexp [4]) show that peer-reviewed experimental spectral *lists* with DOI traceability are in scope for this journal. Concurrent literature corpora that include IR among other modalities (for example NMRSpec/NMRTrans [5]) further motivate an IR-focused, redistributable band-list resource rather than another closed spectrum archive.

Relation to NMRexp and other peers.

NMRexp is the natural comparator: ∼3.3 million experimental NMR records mined from supporting-information PDFs, with expert-scale manual checks and replicate consistency metrics [4]. IRexp is *not* an NMR database, does not claim size superiority, and is orders of magnitude smaller. Its contribution is complementary: redistributable *IR band lists* (cm^{-1} positions only) from PMC Open Access full text plus a Chemotion ELN deposit, with per-record licence pools suitable for commercial vs non-commercial reuse. Absorbance-curve libraries (NIST, SDBS) remain the right choice when full digitised spectra are required; computational IR–NMR sets remain the right choice when simulated multimodal coverage is required [3].

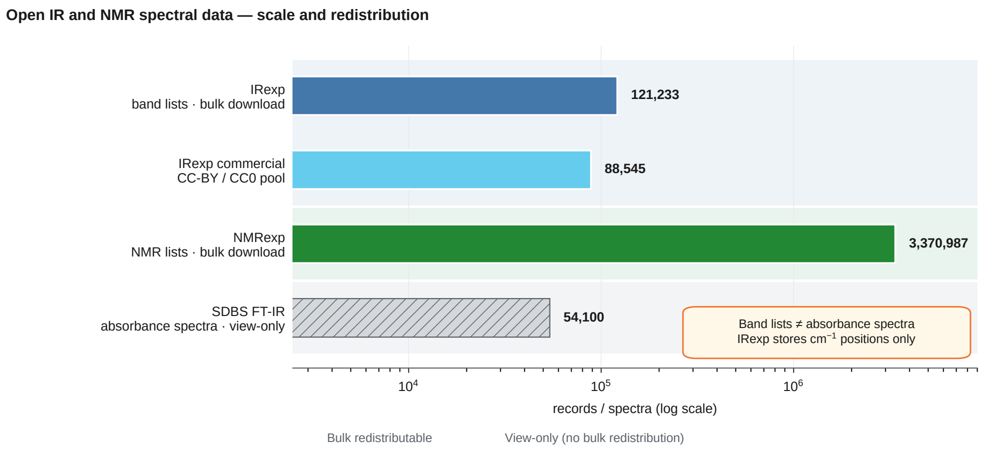

Fig. 1 Positioning of IRexp among open spectral data resources (log scale). **IRexp** holds 121,233 experimental *band lists* (cm^{-1} peak positions) with bulk redistribution and per-record licence pools; the commercial pool (88,545 CC-BY/CC0 rows) is the Zenodo / Sci Data primary artifact. **NMRexp** [4] is the modality peer for literature-mined peak lists (∼3.37 million NMR records; different object). **SDBS** FT-IR holds ∼54,000 digitised absorbance spectra (AIST introduction, 2015) but is view-only with no bulk export. IRexp does not claim absorbance-spectrum coverage or NMRexp-scale size.

In 2025–26, spectroscopic AI agents and autonomous chemistry workflows increasingly consume *structured* experimental peak lists as tool inputs and retrieval substrates — not paywalled PDF prose and not view-only web UIs. Agents that plan characterisation, call spectrum tools, or train IR→structure models need redistributable numeric lists with DOI attribution and explicit licence pools (commercial vs non-commercial vs ShareAlike). IRexp is designed as that substrate: training / retrieval / tool-input material for literature-grounded spectroscopic agents, without embedding diagnosis benchmarks or model leaderboards in this Descriptor.

IRexp fills a specific wedge. Experimental sections of chemistry papers conventionally report per-compound IR band lists (wavenumbers in cm^{-1}) together with $^{1}H/^{13}C$ NMR shift lists. That textual convention is the object language models and many elucidation pipelines consume, and it is a different object from a digitised spectrum. IRexp therefore:

1. Harvests open-access full text from the PMC Open Access Subset bulk S3 mirror and peak lists from the Chemotion FT-IR deposit.
2. Extracts deterministic numeric IR band lists (and co-reported NMR strings where present).
3. Resolves compound names to canonical structures with OPSIN [6], RDKit [7], and SELFIES [8] where possible.
4. Releases extracted numbers only — no PDFs, figures, or article full text — with source accessions for attribution.

Among openly redistributable *text-derived IR band lists*, IRexp is large by record count (121,233). It does not claim to replace SDBS or NIST absorbance libraries; SDBS alone holds more structure-linked *spectra* than IRexp holds structure-linked band lists. The scientific contribution of this Descriptor is the curated dataset, harvest provenance, licence segregation, and validation artefacts — not elucidation accuracy claims. The intended reuse is multimodal pretraining, supervised IR→structure modelling on `irexp_resolved`, and as the literature substrate for complementary elucidation benchmarks described elsewhere (forthcoming ICLR manuscript on IRSpectra-Bench; cite that work for protocol and model results — not reproduced here) [9].

What this Data Descriptor does not contain.

No hypothesis tests, no large-language-model accuracy tables, and no stage-decomposition of elucidation performance. Those belong in the complementary research paper.

2 Methods

2.1 Source corpora

PMC Open Access Subset.

Primary harvest uses the NCBI PMC OA bulk distribution on Amazon S3 (`s3://pmc-oa-opendata`, HTTPS endpoint `https://pmc-oa-opendata.s3.amazonaws.com`) [10]. Harvest window (recoverable from git snapshots): the bulk S3 IR crawl and `seen_papers` / `ir_harvest_snapshot` artefacts were produced on 2026-06-07 (UTC) (`scripts/s3_ir_harvest.py`; incremental auto-snapshots that day through ~134,893 raw IR rows). Chemotion was ingested the same day. A harvest snapshot records 188,016 distinct PMC identifiers scanned (`data/irexp/seen_papers.txt.gz`). The released corpus retains records from 15,416 unique `PMC:*` accessions. Extraction operates on open-access plain text objects at flat S3 keys `PMC{id}.{v}/PMC{id}.{v}.txt`. Europe PMC full-text XML is used for some validation re-fetches. The released IRexp DOIs are exclusively `PMC:*` accessions or the Chemotion deposit DOI; no paywalled publisher DOI appears in the 121,233-record file.

Discovery vs `oa_comm` / `oa_noncomm` packages.

Identifier discovery used NCBI E-utilities `esearch` over PMC with an IR-characterisation query and `open access[filter]`, then fetched matching IDs from the flat S3 text layout (`scripts/s3_ir_harvest.py`). The harvest therefore did not pre-filter by walking the PMC OA Subset's `oa_comm` vs `oa_noncomm` vs other package directories. PMC OA is still not a single licence [10]. Licence truth for redistribution is applied post-hoc: every unique PMCID in IRexp (15,416) was joined to Europe PMC `license` metadata (`scripts/join_pmc_licences.py`); each record carries `license` / `license_pool`. A second-pass recovery (`scripts/apply_crossref_licence_recovery.py`) promoted 1,818 previously empty/unknown rows into commercial, non-commercial, or other pools when Crossref exposed an explicit Creative Commons or ACS AuthorChoice CC URL, or when a fresh Europe PMC query returned a licence string; Elsevier/Springer TDM-only Crossref links were ignored. Commercial-use (CC-BY/CC0) rows form the Zenodo / Sci Data primary pool; NC* are held aside; remaining empty/unknown are excluded from the commercial deposit — aligning redistributable intent with the OA commercial/non-commercial distinction via article-level licences rather than S3 package paths (`data/NOTICE`; `docs/scientific_data/LICENCE_REMEDIATION.md`).

Reproducibility of discovery.

Re-running live `esearch` will drift as PMC grows. The frozen discovery set for this release is `data/irexp/seen_papers.txt.gz` (188,016 PMC identifiers); the curated release retains 15,416 of those accessions. Third parties should treat `seen_papers` + the released JSONL as the reproducible snapshot, not a fresh API crawl.

Chemotion / RADAR4Chem.

1,888 records come from the Chemotion Repository FT-IR collection deposited at RADAR4Chem (DOI `10.22000/OGoEQGlsZGElrgst`) [11], licensed CC-BY-SA-4.0. Ingest (`scripts/chemotion_to_irexp.py`, 2026-06-07): download the MD5-verified deposit; for each of 2,116 ATR-IR analyses, flatten the Quill-delta `content` field to plain text; parse an author-curated IR band list with the same regex extractor and quality gates as the PMC path; resolve the deposit's canonical SMILES with RDKit → InChIKey + SELFIES; keep the richest band list per InChIKey; drop the 8 molecules already present in the PMC pool → +1,888 new structure-resolved rows. All

1,888 Chemotion rows are structure-linked (`has_structure=true`; `inchikey` equals the record `id`). These are not algorithmic peak-picks from absorbance curves; they are author-entered experimental band lists from the ELN deposit, denser than typical paper prose (median 39 bands vs 9 for PMC). Rows carry `license=CC-BY-SA-4.0`, `source=Chemotion`, and `source_doi` equal to the deposit DOI.

Excluded.

AIST SDBS (view-only; no bulk export) [2]; NIST WebBook join code exists in the repository but contributes 0 records to the released IRexp (`ir_source` is always `experimental` for released rows).

2.2 Discovery and fetch (released corpus)

1. Discover IR-reporting OA PMCIDs via NCBI `esearch` (`open access[filter]` + characterisation-format IR query; year/month sliced).
2. Fetch OA full text from PMC-OA S3 plain-text objects (`PMC{id}.{v}.txt`).
3. Ingest Chemotion deposit author-curated band lists + structures (`scripts/chemotion_to_irexp.py`).
4. Persist harvest provenance in `seen_papers.txt.gz` and optional rawer rows in `ir_harvest_snapshot.jsonl.gz` (134,893 rows including intermediate prose fields; the curated release is `irexp.jsonl.gz`).

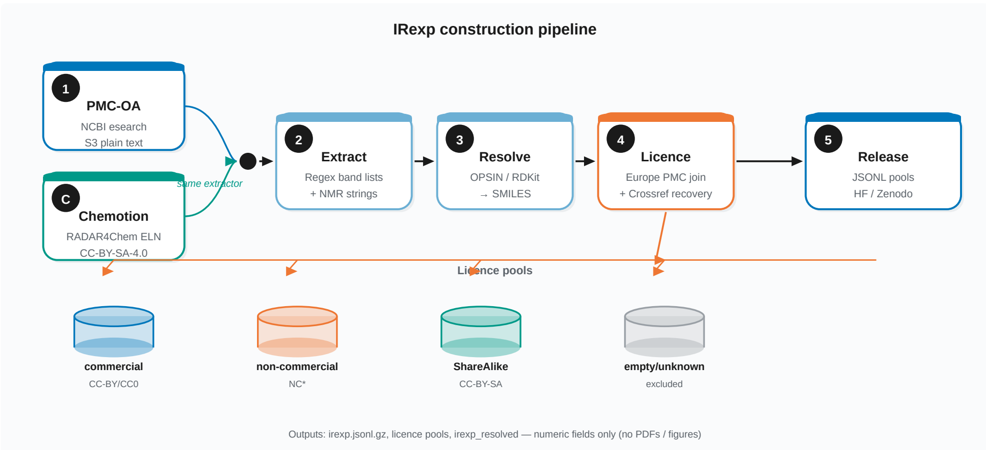

Fig. 2 IRexp construction pipeline for the released corpus. **PMC path:** IR-reporting open-access PMCIDs are discovered via NCBI `esearch`, full text is fetched from the PMC Open Access Subset S3 plain-text mirror, and band lists (plus co-reported NMR strings) are extracted with a deterministic regex pipeline. **Chemotion path:** author-curated ELN band lists and structures from the RADAR4Chem deposit are ingested with the same extractor. Structures are resolved with OPSIN/RDKit where possible; every row is stamped with Europe PMC licence metadata (plus conservative Crossref recovery of empty licences) and split into redistribution pools. Only extracted numeric fields and identifiers are released.

Non-OA / Scrapling fence.

Development adapters for ChemRxiv, Beilstein, and generic publisher pages exist for exploration. They are outside the construction path of the released dataset. No released `source_doi` is a paywalled publisher DOI. Methods for IRexp as published here cite only PMC-OA S3 + Chemotion.

2.3 Extraction (band lists, not spectra)

A deterministic regular-expression pipeline (`spectro_scraper/extract.py`) segments experimental text per compound and extracts:

- IR wavenumbers → `ir_bands_cm-1` (list of floats, cm^{-1});
- ^{1}H and ^{13}C payloads → `h_nmr` / `c_nmr` (author strings, when present).

Gates reject scan-range artefacts and common prose false positives. No curve digitisation is performed: figures are not traced; JCAMP-DX is not stored.

2.4 Structure resolution

Where an IUPAC or systematic name is available, names are converted with OPSIN [6], canonicalised with RDKit [7], and encoded as InChIKey and SELFIES [8]. An optional PubChem [12] name fallback (`USE_PUBCHEM`) handles trivial names. Name→structure failures are expected for trade names, mixtures, and non-IUPAC prose; unresolved rows keep `has_structure=false` and null structure fields. Structure coverage of the full release is 35.5% (43,060 / 121,233). The structure-complete split is shipped as `irexp_resolved`.

2.5 Licence handling

Table 1 Licence pools stamped on the released IRexp corpus.

Pool	Records	Stamp
commercial (CC-BY + CC0)	88,545	`license_pool=commercial`; Zenodo / Sci Data primary
non_commercial (CC-BY-NC*)	21,823	held aside
sharealike (Chemotion + rare PMC SA)	1,897	CC-BY-SA / CC-BY-SA-4.0
empty_unknown	8,963	excluded from commercial deposit
other (CC-BY-ND)	5	held aside

`scripts/join_pmc_licences.py` stamps every row; `scripts/apply_crossref_licence_recovery.py` recovers empty licences when Crossref/EPMC supply explicit CC terms; `scripts/split_license_pools.py` reports provenance and materialises pool files under `data/irexp/licence_pools/`. Narrative and policy: `docs/scientific_data/LICENCE_REMEDIATION.md`.

2.6 Quality tooling

Optional physics gates live in spectro_scraper/quality.py (^{13}C peak count $\leq$ carbon count; ^{1}H integration vs formula; IR wavenumber windows). They were not applied as a hard filter at harvest; Technical Validation reports a full-corpus post-hoc quarantine on irexp_resolved (scripts/quarantine_structure_nmr.py $\rightarrow$ data/audit/structure_nmr_quarantine.jsonl.gz). Transcription and recall-proxy scripts: scripts/audit_extraction.py, scripts/audit_extraction_recall.py.

3 Data Records

3.1 Object definition

Each IRexp record is a JSON object. Required chemistry fields for an IR entry:

Table 2 Core fields of an IRexp JSON record.

Field	Type	Description
id	string	Stable record id (InChIKey when resolved, else internal)
ir_bands_cm-1	float list	Experimental IR peak positions (cm^{-1})
ir_source	string	experimental for all released rows
source_doi	string	PMC:<pmcid> or Chemotion deposit DOI
pmcid	string/null	PMC accession when PMC-sourced
h_nmr / c_nmr	string/null	Author-reported NMR shift strings (not parsed peak tables)
smiles / selfies / inchikey	string/null	Resolved structure encodings
has_structure	bool	True when SMILES present
license / license_pool	string	Per-article stamp (Europe PMC join or Chemotion)
license_raw / license_source	string	Upstream licence token + join provenance
source	string	Chemotion rows only (Chemotion)

Frozen field definitions and validation counts: docs/scientific_data/qc_structure_nmr.json. Not included: absorbance traces, intensities, instrument metadata beyond what appears in source text, PDFs, figures, or full article bodies.

3.2 Files and counts

Paths relative to the project repository / Hugging Face mirror.

Composition of irexp.jsonl.gz.

Licence-pool counts (confirmed Track 1 join, 2026-08-26).

Lookup over 15,416 unique PMCIDs; stamped pool files under data/irexp/licence_pools/. Sum of pools = 121,233 (unchanged).

Table 3 Released IRexp files and record counts.

File	Records	Description
`data/irexp/irexp.jsonl.gz`	121,233	Full curated release
`data/irexp_resolved/irexp_resolved.jsonl.gz`	43,060	100% structure-linked
`data/irexp_release/train_no_bench.jsonl.gz`	42,808	Resolved minus IRSpectra-Bench InChIKeys
`data/irexp_release/train_no_bench_nmr.jsonl.gz`	32,949	Same with both ^{1}H and ^{13}C
`data/irexp_release/pretrain_ir.jsonl.gz`	119,345	PMC-only IR pretrain pool
`data/irexp/seen_papers.txt.gz`	188,016 lines	PMC IDs scanned at harvest
`data/irexp/ir_harvest_snapshot.jsonl.gz`	134,893	Intermediate harvest snapshot
`licence_pools/irexp_commercial.jsonl.gz`	88,545	CC-BY + CC0 (Zenodo/Sci Data primary)
`licence_pools/irexp_non_commercial.jsonl.gz`	21,823	NC* held aside
`licence_pools/irexp_sharealike.jsonl.gz`	1,897	Chemotion + rare PMC SA
`licence_pools/irexp_empty_unknown.jsonl.gz`	8,963	Excluded from commercial

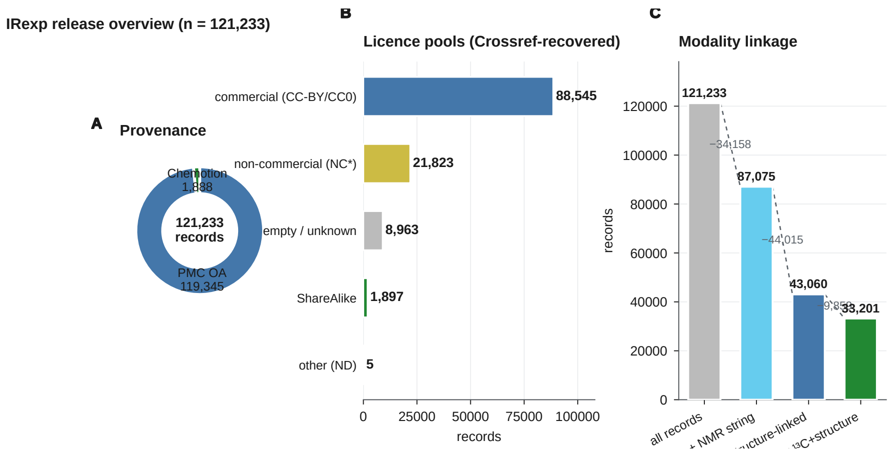

Fig. 3 IRexp release overview from frozen counts. **(a)** Provenance (PMC Open Access Subset vs Chemotion/RADAR4Chem). **(b)** Licence pools after Europe PMC joins and Crossref empty-licence recovery (commercial pool is the Sci Data / Zenodo primary artifact). **(c)** Composition cascade from all band-list records to full IR + ^{1}H + ^{13}C + structure quadruples. Numeric values match Tables 1–5.

The full `irexp.jsonl.gz` remains multi-licence on disk; commercial redistribution must use the commercial pool (or `license_pool == "commercial"`). Hugging Face was remirrored with Crossref-recovered pools and honest card text (`scripts/publish_hf.py`, 2026-08-27; commercial 88,545).

Median bands: 9 (PMC), 39 (Chemotion). All 1,360,866 released IR band values fall inside 350–4000 cm^{-1} (full-corpus range check; Technical Validation).

Table 4 Composition of the full curated release.

Slice	Count
All IR band-list records	121,233
With ^{1}H and/or ^{13}C NMR	87,075 (72%)
With resolved structure	43,060 (35.5%)
Structure + any NMR	40,702
Full IR + ^{1}H + ^{13}C + structure	33,201
PMC OA provenance	119,345
Chemotion provenance	1,888
Unique PMC accessions	15,416

Table 5 Licence-pool file counts (Track 1 join).

Pool file	Count	Notes
`irexp_commercial.jsonl.gz`	88,545	CC-BY / CC0 — Zenodo primary
`irexp_non_commercial.jsonl.gz`	21,823	CC-BY-NC* held aside
`irexp_empty_unknown.jsonl.gz`	8,963	empty / unresolved — excluded from commercial deposit
`irexp_other.jsonl.gz`	5	CC-BY-ND (Crossref recovery)
`irexp_sharealike.jsonl.gz`	1,897	Chemotion CC-BY-SA-4.0 (1,888) + rare PMC SA

3.3 Access

- Hugging Face: https://huggingface.co/datasets/ilkhamfy/IRexp
- GitHub development mirror: https://github.com/IlkhamFY/spectro-agent
- Zenodo archival snapshot: DOI pending mint after honest multi-licence metadata (data-only deposit; commercial pool primary — not the combined IRSpectra-Bench research archive).

4 Technical Validation

Machine-readable package: `docs/scientific_data/qc_structure_nmr.json` (and artefacts under `data/audit/`; methods note `docs/scientific_data/TV_AND_SCALE_PASS.md`). Numbers below are seed-fixed descriptive checks; Wilson 95% intervals are reported for the expanded recall proxy and chemist-proxy pass rates. **All checks below are automated** unless marked otherwise — they do **not** claim NMRexp-parity human expert audits.

4.1 Transcription fidelity (PMC)

On a seed-fixed random sample of 60 PMC-sourced records (`scripts/audit_extraction.py --n 60 --seed 0`), each article was re-fetched from Europe PMC and every recorded wavenumber was checked against the source text ($\pm 1\,\mathrm{cm}^{-1}$ integer match). Result: 560/560 bands and

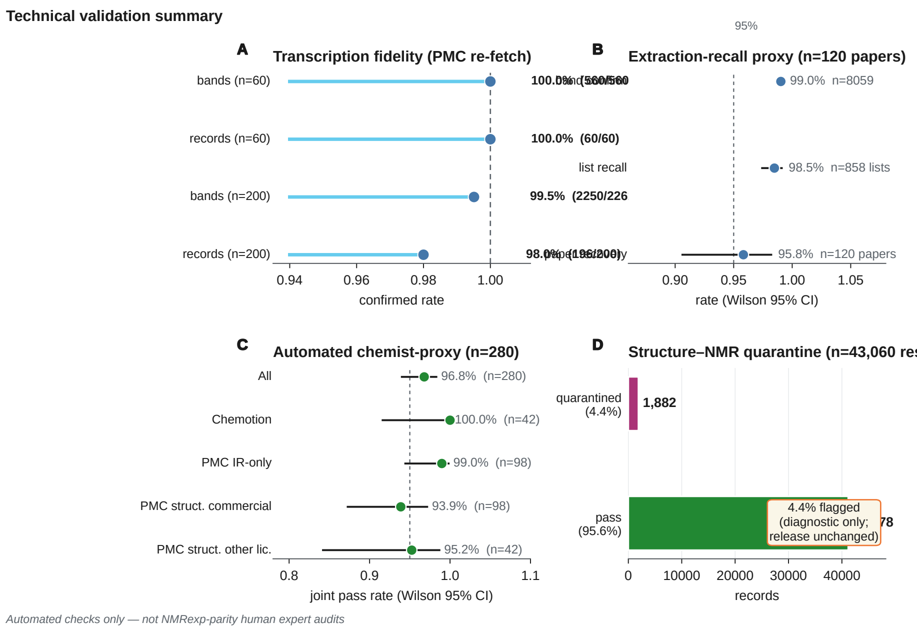

Fig. 4 Technical validation summary (automated checks only). **(a)** PMC transcription fidelity on Europe PMC re-fetch ($n = 60$ and $n = 200$; $\pm 1\,\mathrm{cm}^{-1}$ integer match). **(b)** Harvest-path extraction-recall proxy on 120 PMC papers (band confirmation, list-level recall, paper-level recovery; Wilson 95% CI). **(c)** Stratified automated chemist-proxy ($n = 280$; joint multi-check pass rate by stratum). **(d)** Full-corpus structure–NMR quarantine on `irexp_resolved` (43,060 rows; diagnostic flags only). These metrics bound automated fidelity and consistency; they are not human molecular-skeleton audits of the kind reported for NMRexp.

60/60 records confirmed. Enlarged sample $n = 200$ (same script/seed family, `data/audit/extraction_audit_n200.json`): 2,250/2,261 bands (99.51%) and 196/200 records (98.0%) fully confirmed. The four incomplete records (PMCIDs 5713685, 4189708, 12202350, 6259152) missed 11 bands total on re-fetch match — consistent with plain-text / formatting drift between harvest and Europe PMC re-fetch, not wholesale hallucination. This bounds automated transcription error (mis-parsed or unit-mangled values); it is not a human PDF mark-up of the kind reported for NMRexp [4].

4.2 Extraction-recall proxy (PMC, harvest path)

A human mark-up of every IR string in every paper remains the gold standard. As an automatic proxy on the actual harvest path (`scripts/audit_extraction_recall.py --n 120 --seed 0`): re-fetch PMC-OA S3 plain text for 120 distinct source PMCIDs; re-run `extract_records`; compare band-sets to released rows ($\pm 1\,\mathrm{cm}^{-1}$). Result (`data/audit/extraction_recall_proxy_n120.json`): 7,981/8,059 released bands confirmed in re-extract (rate 0.9903; Wilson 95% CI [0.9879, 0.9922]); 845/858 released IR lists recovered (list-level recall proxy 0.9848; Wilson 95% CI [0.9743, 0.9911]); 115/120 papers recovered every released list (0.9583; Wilson 95% CI [0.9062, 0.9821]).

Of 871 re-extracted IR lists, 811 matched a released list (0.9311); 18/120 papers yielded *extra* re-extracted lists relative to the curated release (parser over-fire vs post-harvest dedup/curation, not missed released compounds). A smaller prior $n = 40$ run remains archived as `data/audit/extraction_recall_proxy.json`. This is not a substitute for expert human recall.

4.3 Stratified chemist-proxy audit (automated)

To approach NMRexp's $n \approx 300$ audit *sample size* without fabricating chemist signatures, we ran a seed-fixed stratified multi-check **automated chemist-proxy** (`scripts/audit_chemist_proxy.py --n 280 --seed 0`; `data/audit/chemist_proxy_audit.json`). Strata (quotas): PMC structure-linked commercial (98), PMC structure-linked other licence (42), PMC IR-only (98), Chemotion (42). Checks: IR window and band-list sanity; RDKit formula vs ^{13}C peak count / ^{1}H integral physics on structure-linked rows; PMC Europe PMC + S3 transcription re-confirm ($\pm 1\,\mathrm{cm}^{-1}$).

Table 6 Automated chemist-proxy audit ($n = 280$, seed 0). Not a human skeleton audit.

Stratum / metric	n	Pass / rate	Wilson 95%
All strata (joint automated checks)	280	271 / 0.9679	[0.9401, 0.9830]
Chemotion	42	42 / 1.000	[0.9162, 1.000]
PMC IR-only	98	97 / 0.9898	[0.9444, 0.9982]
PMC structure commercial	98	92 / 0.9388	[0.8728, 0.9716]
PMC structure other licence	42	40 / 0.9524	[0.8421, 0.9868]
Transcription bands (PMC refetch)	2,508	2,500 / 0.9968	—
Transcription records fully confirmed	238	236 / 0.9916	—
Structure physics pass (RDKit formula gates)	182	177 / 0.9725	—

Dominant fail modes in-sample: ^{1}H integral > formula+2 (4), duplicate integer bands (2), transcription miss (2), ^{13}C peaks > carbons (1). The 97.25% structure-physics pass rate on 182 structure-linked rows is a **chemist-proxy consistency** metric, not NMRexp's human molecular-skeleton correctness (∼98% on $n \approx 300$ manual checks).

4.4 Structure–NMR consistency and quarantine (resolved)

Sample (prior).

On 500 `irexp_resolved` records with ^{13}C text (seed 0): 17/500 (3.4%) listed more peaks than carbons. On 500 with ^{1}H text: integrals > formula H+2 in 17/497 (3.4%).

Full resolved corpus.

`scripts/quarantine_structure_nmr.py` applied the same physics gates to all 43,060 structure-linked rows. 1,882 (4.37%) fail ≥1 hard check and are listed in `data/audit/structure_nmr_quarantine.jsonl.gz` (diagnostic only — release files unchanged). Among rows with the relevant modality: ^{13}C peaks > carbons 1,194/34,231 (3.49%); ^{1}H integral > formula+2 1,141/39,672 (2.88%); IR out-of-range 0; unparseable SMILES 0. Sample rates and full-corpus rates agree closely, supporting stability of the gates. Re-users should filter quarantined IDs before supervised training (default recommendation: drop the quarantine list unless a noisier corpus is intentionally desired). These rates are integrity diagnostics, not an expert skeleton audit (NMRexp reported ∼98% skeleton correctness on a manual $n \approx 300$ sample [4]; that human check remains pending for IRexp).

4.5 IR physical window

Every band in the full 121,233-record release lies in $[350, 4000]\,\mathrm{cm}^{-1}$ (0 / 1,360,866 out of range). This is a necessary range gate only: IRexp does not store intensities, solvents, or ATR vs transmission metadata.

4.6 QC status

Table 7 Technical validation checklist.

Check	Status
Per-PMCID licence join + Crossref empty recovery	Done — commercial 88,545
Transcription fidelity n=60 and n=200	Done (automated re-fetch)
Extraction-recall automatic proxy (n=120 papers)	Done (human recall still optional)
Stratified chemist-proxy audit (n=280)	Done (automated; not human expert)
Full-corpus structure–NMR quarantine	Done — 1,882 / 43,060 flagged
Expert human structure spot-check ($n \geq 100$)	Deferred (human)
NMRexp-style replicate MAE for IR lists	Not applicable / not claimed

5 Usage Notes

- **Band lists ≠ spectra.** Do not evaluate models trained on IRexp as if they had seen full absorbance curves.
- **AI agents / LLM tool-use.** Prefer the JSON band-list fields and `source_doi` as tool returns or retrieval hits; do not scrape paywalled PDFs or view-only SDBS pages when a redistributable pool already exists. Productized agents must filter by `license_pool` (commercial vs NC* vs ShareAlike) rather than treating the full dump as a single licence.
- **Separate pools by density and licence.** PMC (sparse, author-transcribed) vs Chemotion (denser, author-curated ELN band lists). Higher median band count

does not imply more complete vibrational assignment — Chemotion lists are author-entered, not peak-picked curves. Combined redistribution of Chemotion-derived rows must honour CC-BY-SA-4.0 (ShareAlike); do not relicence SA rows as CC-BY.

- **Do not assume PMC = CC-BY-4.0.** Filter to `license_pool == "commercial"` (or use `irexp_commercial.jsonl.gz`) for commercial redistribution; attribute via `source_doi` / `pmcid`.
- **Structure–NMR quarantine.** Before supervised training on `irexp_resolved`, drop IDs in `data/audit/structure_nmr_quarantine.jsonl.gz` by default ($\sim$4.4% of resolved rows) unless a noisier training set is intentional.
- **Training without benchmark leakage.** If using the complementary IRSpectra-Bench problems (companion research paper), fine-tune from `train_no_bench.jsonl.gz` (or rebuild with `scripts/build_train_no_bench.py`) so benchmark InChIKeys are withheld. Protocol and model results live only in that companion manuscript.
- **Structure coverage.** Prefer `irexp_resolved` for supervised structure tasks; 64.5% of records lack SMILES.
- **FAIR reuse example.** Load only `license_pool == "commercial"` rows for a redistributable training set; drop quarantine IDs; keep `source_doi` / `pmcid` for attribution; prefer `irexp_resolved` when SMILES is required. ShareAlike Chemotion rows may be remixed only under CC-BY-SA terms.
- **Attribution.** Cite this Data Descriptor / Zenodo DOI (when minted) and attribute originating articles through each record's `source_doi`.

5.1 Limitations

- **Object.** IRexp is a band-list corpus, not an absorbance-spectrum library and not an NMR resource comparable to NMRexp in scale or annotation richness.
- **Technical Validation depth.** Automated transcription ($n = 200$), harvest-path recall proxies ($n = 120$ papers with Wilson intervals), and a stratified chemist-proxy ($n = 280$) close the *sample-size* gap versus NMRexp's $n \approx 300$ audits, but remain machine checks. They are weaker evidence than NMRexp's manual PDF mark-up and replicate MAE analyses. No human molecular-skeleton audit has been completed for IRexp.
- **Metadata sparsity.** Intensities, solvents, and instrument modes are generally absent; the IR window check is necessary but narrow.
- **Structure coverage and name resolution.** Only 35.5% of records are structure-linked; OPSIN/PubChem failures leave many IR lists without SMILES.
- **Licence mix.** The full `irexp.jsonl.gz` is multi-licence; commercial Zenodo/Sci Data redistribution is the commercial pool only (88,545 after Crossref recovery of previously empty rows).
- **Scope of this paper.** Elucidation benchmarks and model results are out of scope for this Data Descriptor; cite the companion research manuscript (IRSpectra-Bench / ICLR track) for those claims. Do not treat this Descriptor and the companion paper as interchangeable publications of the same results.

6 Data Availability

IRexp numeric extracts are available at:

- Hugging Face Datasets: https://huggingface.co/datasets/ilkhamfy/IRexp
- GitHub: https://github.com/IlkhamFY/spectro-agent (`data/irexp/`, `data/irexp_resolved/`, `data/irexp_release/`)
- Zenodo: archival DOI at proof for a *data-only* deposit (primary artifact = commercial pool, `cc-by-4.0` metadata + SA companion). Do not reuse the combined IRSpectra-Bench research `.zenodo.json` as the Sci Data archival record.

Licensing summary (honest):

- Chemotion (1,888): CC-BY-SA-4.0 [11].
- PMC (119,345): mixed Creative Commons — stamped per article; commercial redistributable 88,545 (CC-BY/CC0); NC* 21,823 held aside; empty/unknown 8,963 excluded from commercial Zenodo (`LICENCE_REMEDIATION.md`).
- Only extracted numeric fields and identifiers are redistributed; source full texts are not.

7 Code Availability

Harvesting, extraction, structure resolution, licence pool splitting, and validation scripts are in https://github.com/IlkhamFY/spectro-agent under the MIT License (`LICENSE`). Principal modules: `spectro_scraper/` (`extract.py`, `normalize.py`, `pipeline.py`, `quality.py`); release harvest `scripts/s3_ir_harvest.py`, `scripts/chemotion_to_irexp.py`; licence join `scripts/join_pmc_licences.py`, `scripts/apply_crossref_licence_recovery.py`; validation `scripts/audit_extraction.py`, `scripts/audit_extraction_recall.py`, `scripts/audit_chemist_proxy.py`, `scripts/quarantine_structure_nmr.py`; pool split `scripts/split_license_pools.py`. The version corresponding to this Descriptor will be tagged at Zenodo deposit time.

Author contributions

I.Y.: conceptualization, methodology, software, data curation, validation, writing (original draft).
R.A.V.-H.: conceptualization, supervision, writing (review and editing).

Competing interests

The authors declare no competing interests.

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